

MINI-PROJECT 1 — NYC TAXI DATA ANALYSIS

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Roll Number: 2305122

Seed: 5122

KEY FINDINGS

The dataset contained 19,566 raw rows, including 291 exact duplicate rows and 196 rows with NULL fare_amount. After cleaning, 19,002 rows remained. The total fare was \$233,671.02, with an average fare of \$12.30, average trip distance of 2.852 miles, and average trip duration of 15.209 minutes. The busiest pickup hour was 19:00 with 1,250 trips, while Friday had the most trips. The top pickup ZIP code was 10001 with 1,060 trips. Thursday had the highest average fare at \$12.66 across 2,781 trips. The longest trip was 30.6 miles with a \$95 fare from ZIP 11371 to 7114, while the most expensive trip was \$275 for a 20.85-mile trip from ZIP 10013 to 7008.

DATA CLEANING

Exact duplicates were removed because they can count the same trip more than once. NULL fare_amount rows were removed because fare is required for fare-based analysis, while non-positive fares and non-positive trip distances were removed because they are invalid for normal taxi trips. Rows where the dropoff time was earlier than or equal to the pickup time were also removed because they have invalid trip duration. The NULL fare rule is arguable because missing fares could potentially be recovered from another reliable source before dropping the rows.

PART D — OWN QUESTION

Question: Which day of the week has the highest average fare? Thursday had the highest average fare at \$12.66 across 2,781 trips. A taxi company would care because identifying days with higher average fares can help with revenue planning and driver allocation.

PART E — SUBMISSION SIGNATURE

===== MINI-PROJECT 1 SIGNATURE =====

SEED : 5122

CLEAN ROWS : 19002

DATASET SIGNATURE : 9488604923

TOTAL FARE : 233671.02

AVG DISTANCE : 2.8524

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